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(1) Research Program/Project/Study Title: Acceptability Milkfish Belly Longganisa	
Proponent Name:	LOUELLA P. DADIVAS
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	JANET LEDESMA ASSI Prof.3
Campus/Satellite College/Address:	CAPSU DAYAO SATELLITE COLLOGE
Telephone/Cell Phone/Fax Number:	(036)620-2647
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	ROXAS CITY
		Province:	CAPIZ
		Region:	VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Food Engineering	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Longganisa, a traditional Filipino sausage, has gained significant popularity in Filipino cuisine. However, concerns about the environmental impact and health implications of meat-based longganisa have emerged due to its high demand. This study aims to explore the acceptability of milkfish belly as an alternative ingredient for longganisa production, addressing these concerns while preserving the cultural and sensory aspects of this beloved dish.

The research was conducted at Capiz State University, Dayao Satellite College, Roxas City, with a total of one hundred evaluators participating in the study. The evaluators rated the four formulations based on their sensory attributes. The data analysis revealed that Formulation 4, which consisted of 100% milkfish belly, was the most preferred formulation with a general acceptability rating of "like very much." Furthermore, when comparing the most preferred formulation to the sensory characteristics of the control formulation, it was found that Formulation 4 (100% milkfish belly) is acceptable as a meat substitute. The study also examined consumer acceptability, categorizing the target population into "Child," "Teen," and "Adult" groups. The results indicated that these groups described Formulation 4 as "Like Very Much."

Overall, this research demonstrates that milkfish belly longganisa, specifically Formulation 4, is highly acceptable to consumers as a meat substitute. The findings suggest that milkfish belly has the potential to be a healthier and more affordable alternative to pork meat in longganisa production, contributing to sustainable and nutritious food choices in Filipino cuisine.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

, Longganisa, Milkfish

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The use of meat substitutes has gained considerable attention in recent years, driven by concerns over environmental sustainability, health consciousness, and changing dietary preferences. Longganisa, a traditional Filipino sausage, holds a significant place in Filipino cuisine. However, the high demand for meat-based longganisa has raised concerns regarding its environmental impact and health implications. According to Mihai Andrei (2022), one of the main reasons why Pork is bad is the composition of its high content of saturated fats and cholesterol that causes diabetes, cardiovascular disease, obesity, cirrhosis, and multiple types of cancer. Exploring alternative ingredients for longganisa production can address these concerns while maintaining the cultural and sensory aspects of this beloved dish.

Milkfish is a popular and widely available fish species in the Philippines, known for its delicate flavor and tender texture. The use of milkfish belly as a potential meat substitute in longganisa presents an intriguing possibility. Milkfish belly offers an opportunity to develop a sustainable and healthier alternative to traditional longganisa, while still capturing the distinctive taste and texture that make this dish a favorite among Filipinos.

The researchers conducted this study and investigated the acceptability of milkfish belly as a meat substitute for longganisa among consumers. The study will determine and evaluate the nutritional analysis, and consumer perceptions to gauge the viability of this alternative ingredient. Understanding the acceptability of milkfish belly longganisa will provide insights into the feasibility of incorporating sustainable and healthier options into traditional Filipino cuisine.

In this research paper, we will explore relevant studies on meat substitutes, the sustainability of fish as a meat alternative, and the potential health benefits of incorporating fish into the diet. By examining the acceptability of milkfish belly as a meat substitute for longganisa, this research aims to contribute to the development of sustainable and healthier options within Filipino cuisine. The findings of this study can inform consumers, food producers, and policymakers about the potential benefits and challenges associated with incorporating alternative ingredients into traditional dishes, fostering a more sustainable and health-conscious food culture

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Materials/Utensils

The following is the list of materials used in this study. Knives, Basin, Weighing scale, Measuring spoon, Trays, Teaspoons, 4x10 Plastic for packaging, Mixing bowl, Tablespoons, Chopping board, Cup, and Measuring cups.

Ingredients

The ingredients used in the preparation of Milkfish belly longganisa are the following: Milkfish belly, water, pineapple juice, anisado wine, TVP, garlic, black pepper, phosphate, curing Salt, MSG, and salt.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Experimental Procedure

The following are the steps to be followed in the preparation of Milkfish belly longganisa:

Preparation of Milkfish belly:

The milkfish deboning process starts with washing the freshly harvested milkfish. it is then split (butterfly cut) and internal organs including the bile are removed. Once this is done, the deboning process follows suit. In the deboning procedure, the spines (dorsal, ventral, lateral, intramuscular spines) and rib bones are removed gradually using sterile mosquito forceps. After deboning, Milkfish belly was separated from the meat and set aside.

Preparation of Milkfish belly longganisa:

- 1. Weigh 1000 grams of Milkfish belly and set aside
- 2. Mix salt and cure salt to Milkfish belly
- 3. Add phosphate dissolved in water and mix very well.
- 4. Mix the other dry ingredients well and add to the Milkfish belly.
- 5. Add the liquid ingredients. 6. Mix well until uniformly blended.
- 7. Cure in the refrigerator for 24 hrs.
- 8. Packed in PP bags and store in the freezer until needed.

Experimental Design and Treatments

This study used the single factor experiment in Completely Randomized Design (CRD) with four (4) treatments and each treatment have two (2) replications. The Analysis of Variance (ANOVA) used to analyze the data. The treatments are as follows: The proportion of the longganisa was divided into the following ratios per 1000g, Formulation 1 which is 75% ground pork + 25% Milkfish belly, formulation 2 which is composed of 50% ground pork + 50% Milkfish belly, formulation 3 with 25% ground pork + 75% Milkfish belly and lastly, formulation 4: 100% milkfish belly and 100% of pork serves as our control group. The other ingredients will be held constant (See table 1)

Data Gathering Procedure

The data gathered using score cards where ratings are indicated in terms of general acceptability of the product, most preferred formulation and a preference test. The evaluators in this study are ten (10) semi-trained panelists from Food Technology students and teachers of CapSU Dayao Satellite College and One Hundred (100) panelists are composed of children (5-12 yrs old), teens (13-19) yrs old), adult (20-39 yrs old) and middle age adult (40-59 yrs old).

Data Analysis Procedures

Statistical Package for Social Sciences (SPSS) will generate and process all the data needed in this study. The data will be analyzed using the mean and analysis of variance (ANOVA). The weighted mean will be used to determine the consumer's acceptability of the Milkfish belly longganisa.

Scoring of Variables

The criteria utilized to evaluate the m general were the 9-Point Hedonic Scale rating used liked extremely- 9, for liked very much- 8, for liked moderately- 7, for liked slightly-6 , for neither liked nor disliked- 5, disliked slightly- 4, for disliked moderately- 3, for disliked very much- 2 and for disliked extremely-1 (Calmorin, 2006)

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Most preferred formulation among treatments.

Table 1 shows the result ranking test of the (4) different treatments Results showed that Treatment 1 (75% ground Pork + 25%Milkfish Belly) had mean of 2.55 followed by Treatment 2 (50% ground Pork + 50% Milkfish Belly) with a mean of 3.05 and Treatment 3 (25% ground Pork + 75% Milkfish Belly) got the mean of 3.05 and Treatment 4 (100% milkfish belly) had a mean of 1.40.

Based on the results Treatment 4(100% milkfish belly) with the mean of 1.40 is the most preferred formulation (MPF) among treatments.

Table 1: Most preferred formulation among treatments.

Treatments	Trial 1	Trial 2	Mean
Treatment 1	2.5	2.2	2.55
Treatment 2	2.8	2.8	3.05
Treatment 3	2.6	2.8	3.05
Treatment 4	1.3	1.9	1.40

Friedman’s Test result among treatments.

Table 2 shows that Friedman's test results of the ranking test of the 4 Treatment at a 5% level of significance show that there is no significant difference in the rank on the four (4) treatment. This qualifies every treatment acceptable as the Most Preferred Treatment (MPF). Treatment 4 was designated as the Most Preferred Treatment MPF for this study based on the researchers' analysis of the mean MPF ranks. Therefore, Treatment 4 with 100% pure belly is accepted as the Most Preferred Treatment (MPF) of this study.

Table 2 Friedman’s Test among treatments.

N	20
Chi-Square	21.271
Df	3
Asymo. Sig	.000

General acceptability of milkfish belly longganisa among treatments.

Table 3 Shows that the result of the acceptability of milkfish belly longganisa, Result showed that Treatment 1 (75% ground Pork + 25% Milkfish Belly) had a mean of 7.30 with the verbal interpretation of like moderately and treatment 2 (50% ground Pork + 50% Milkfish Belly) had a mean of 6.55 like slightly' treatment 3 (25% ground Pork + 75% Milkfish Belly) had a mean of 6.95 like slightly. treatment 4 (100% Milkfish belly) got the highest mean of 8.05 with the verbal interpretation of like very much.

The results show that among treatments, treatment 4 with 100% milkfish belly rated like very much for the most acceptable treatment.”(See table 3)

Table 3: General Acceptability of Milkfish Belly Longganisa

Treatments	Trial 1	Trial 2	Mean	Verbal interpretation
Treatment 1	7.7	8.4	7.30	Like Moderately
Treatment 2	6.8	6.3	6.55	Like Slightly
Treatment 3	6.9	7.0	6.59	Like Slightly
Treatment 4	7.4	7.2	8.05	Like very much
Total			7.21	

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Result of the ANOVA test of the general acceptability of the four (4) treatments..

Table 4 the ANOVA result in general acceptability shows that there is no significant difference in the acceptability of the four (4) treatments. With a confidence level of 5%, the result of the Friedman Test shows that there is no significant difference in treatment 1,2,3, and 4 evaluated as like moderately, like slightly and like very much.

Table 4. Result of the ANOVA test of the general acceptability of the four (4) treatments

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	24.338	3	8.113	5.267	.002
Within Groups	117.050	76	1.540		
Total	141.388	79			

To Compare the Overall Sensory Characteristics of Most preferred formulation with that of the Control.

Table 5 shows the result for the paired preference test of the most preferred formulation (MPF) vs. Control Trials. MPF got the highest based on the 10 trained panelists. Using Statistical Table (Control is pure 100% pork) had a mean of 2.5 and the (MPF 100% Milkfish belly) had a mean of 7.5 Therefore the most preferred formulation (MPF) is the 100% Milkfish Belly since it got the highest mean compared to Control which is 100% Pure Meat. The Paired Preference test showed that there was no significant difference between the control and most preferred formulation of Milkfish Belly Longganisa which makes the 100% milkfish Belly acceptable as a meat substitute.

Table 5: Most Preferred Formulation(MPF) Vs. Control To Compare the Overall Sensory of the Control.

Table 5. Results of overall sensory characteristics of Most preferred formulation with that of the Control.

Treatments	Trial 1	Trial 2	Mean
MPF 1	6	9	7.5
CONTROL 2	4	1	2.5

Consumer General Acceptability of MPF of Milkfish Belly Longanissa

Table 6 shows the consumer test for general acceptability of the Most Preferred Formulation (MPF) of Milkfish Belly Longganisa shown in Table 6 shows that Teens (13-19 yrs old) gained a rating of 8.24 “like very much”, Child (5-12 yrs old) gained rating 8.68 “like very much”, Middle age adult (40-59 yrs old) gained rating 7.48 “like moderately”, Adult (20-39 yrs old) gained rating 8.28 “like very much”. Therefore, based on the gathered data, the target population of the study were the “Child” “Teen” and “Adult” and described the formulation 4 as “Like Very Much”

Table 6 Consumers Test for General Acceptability

	Child 5-12 yrs old	Teen 13-19 yrs old	Adult 20-39 yrs old	Middle age adult 40-59 yrs old
Total	217	206	207	197
Mean	8.68	8.24	8.28	7.88
Description	LVM	LVM	LVM	LM

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